Vojtěch Mikula

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Education

Charles University, Faculty of Mathematics and Physics 2024 - present
Mathematical Modelling, BSc. Prague

- Linear Algebra I-II, Probability, Numerical Methods, Mathematical Analysis I-IV (e.g. calculus of variations, Fourier analysis, distributions), Classical Mechanics, ODEs
- Elective: Intro to AI (Bayesian reasoning, game theory, ML, CSP), currently Introduction to Machine Learning

Human-Aligned AI Summer School (HAAISS) 2026, Prague

Archbishop Grammar School 2016 - 2024, Prague

- 93.8th percentile in national Advanced Mathematics exam
- Ranked first from Czech teams with team *Fyzikalení* at international physics competition Fyziklání 2024 (11th out of 250 teams total, 7th out of 74 in category A)
- 1.0 average in Maturita with subjects: Mathematics, Informatics, English, Czech Literature

Experience

Software Development Intern 2025
Unichargepay

- Implemented Flask backend of a web application enabling customers to browse data about their chargers
- Worked with ssh remote development, prod/test/dev environments, gitea, bookstack

Instructor, scout educational course Trinity 2023 - 2025
Junák – český skaut

- Collaborated in a team of 10 instructors to deliver a course focused on (nonviolent) communication and respectful approach for 30 participants aged 16–21 across 10-day summer camp and 2 weekend modules
- Planned and co-organized events and workshops, taught rhetoric, team debating and outdoor cooking, maintained website and course knowledge base, automatized generation of grocery lists

Projects

Toy Models of Superposition 2026, GitHub

- Read Neel Nanda's guides on starting with Mechanistic Interpretability; studied Anthropic's paper *Toy Models of Superposition*; key finding: models exploit feature sparsity to represent more features than dimensions
- Replicated a figure demonstrating the emergence of superposition with increasing sparsity in small ReLU models

Game tree search 2026

- Implemented alpha-beta pruning with transposition tables for two-player resource collection on a graph; optimized with precomputed distances and terminal state analysis

Science Week at Faculty of Nuclear Sciences and Physical Engineering, CTU 2023, GitHub

- In a team of three students, mentored by Martin Vaněk, implemented a multi-layer perceptron neural network for digits classification; performed hyperparameter tuning to maximize accuracy
- Implemented forward pass, backpropagation, and gradient descent from scratch using only NumPy
- Presented the results at a student conference within Science Week

Skills

Programming:

- Python: NumPy, matplotlib; project level experience with scikit-learn, constraint, PyTorch; rapid prototyping in new libraries
- Wolfram Mathematica: solving various problems involving ordinary differential equations

Environments/Tools: Jupyter, VS Code, Git, LaTeX, Ubuntu
Language: English C2 (Cambridge Advanced exam with grade A), Spanish passive approx. B1, Czech native